

Aaron Chu

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Computer Engineering student at Rutgers University, graduating May 2028

EXPERIENCE

Software Engineering Intern

June 2026 – Aug. 2026

Magnefy (industrial transformer monitoring)

Palo Alto, CA

- Achieved 0.991 macro F1 classifying partial-discharge type (corona, internal, surface) on 27K lab-bench pulses (grouped 5-fold cross-validation) by training and benchmarking **XGBoost**, **logistic regression**, and **MLP** classifiers
- Designed and shipped a rule-based validity screen that replaced an **ML**-trained anomaly gate, moving arcing/digital-noise detection to a pre-processing rule stage on raw samples; achieved 100% arcing recall (695/695 labeled events) at 1.48% false positives as the production screen of record
- Produced unified per-asset health summaries by building asset-monitoring, data-**drift**, and **trend**-history modules from scratch and a Dynamic Health Index encoder that combines baselines and classifier confidence
- Built the automated data-extraction and **ML** inference pipeline over existing **S3** IoT storage: 598 captures (193K waveforms) across 9 units with no manual trigger, using an **S3**→**SQS** event pipeline across 4 **Railway** services

Lead Systems Integration Engineer

Sept. 2025 – Present

University Nanosatellite Program (Phase A satellite mission)

New Brunswick, NJ

- Increased data reliability by building distributed **Python** services on **Linux/Raspberry Pi** (**multithreaded** polling, structured sensor ingestion) as lead of a 10-member systems integration team
- Strengthened multi-sensor data collection on embedded systems by building integration tests and diagnostic tooling for hardware interfaces and analyzing failure cases
- Authored 3 Critical Design Review verification test procedures (communications, attitude determination and control, thermal) for SPICESat, a 6-unit **CubeSat**, tracing 20+ requirements to verification methods and resolving conflicts across interface control documents and vendor datasheets

Drone Support Intern

June 2025 – Aug. 2025

Integem

Cupertino, CA

- Developed and tested **Python** firmware for **ESP32** platforms, writing scripted integration workflows to validate control loops and sensor bring-up
- Integrated computer vision modules (**AprilTag**, optical flow) into an embedded navigation stack, applying algorithmic tradeoffs to enable autonomous flight features
- Built a scalable **IoT** telemetry pipeline streaming GPS/weather data over **Helium** to **InfluxDB/Grafana** for a high-altitude flight experiment

PROJECTS

DeskPartner

github.com/aarochu/DeskPartner

- Architected the 5-module software stack (arm control, vision calibration, **VLM** planner, data collection, training) for a 5-person hackathon team; earned Honorable Mention at the Embodied Metal Hackathon
- Ported **Rerun**'s robotics logging/analysis toolkit (record, catalog, query, replay) to an unsupported 7-joint arm, open-sourced it, and led the team's Rerun bounty integration
- Implemented arm motion primitives, **ArUco**-based camera-to-arm calibration, and a **VLM** perception-to-action loop, validated in dry-run

PocketLab

github.com/ja-zoe/pocket-lab

- Built full-stack **IoT** lab platform (team of 4): **React/TypeScript** frontend, **WebSocket** backend, and **ESP32** edge devices for real-time experiment control
- Implemented real-time data pipelines with **Recharts/Three.js**, auth, offline fallback, and export APIs for classroom deployment

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript/JavaScript, SQL

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy

Backend, Web, and Cloud: AWS, FastAPI, Node.js, PostgreSQL, Supabase, React, Expo, Docker

Tools: Git, Linux, pytest, Streamlit, Pydantic

EDUCATION

Rutgers University

New Brunswick, NJ

BS Computer Engineering

May 2028

- **Coursework:** Software Engineering, Data Structures, **Distributed Deep Learning**, Computer Architecture, Discrete Mathematics, Probability & Random Processes